

## Lecture 2 Web Design and Programming

### Tags in HTML:

In HTML, there are opening tags and closing tags. The characters in the brackets indicate the tag's purpose. For example, in the tags above the **p** stands for paragraph. The closing tag has a forward slash after the < symbol.

</p>

Closing Tag

<p>

Opening Tag

An **Element** comprises the **opening tag** and the **closing tag** and any content that lies between them.

<p>Text</p>

Element

Similarly, an **Attribute** provide additional information about the contents of an element. They appear on the opening tag of the element and are made up of two parts: a name and a value, separated by an equals sign.

<p style="text-align: center;">Text</p>

Attribute

### Basic Tags in HTML

**Heading Tags:** Any document starts with a heading. You can use different sizes for your headings. HTML also has six levels of headings, which use the elements <h1>, <h2>, <h3>, <h4>, <h5>, and <h6>. While displaying any heading, the browser adds one line before and one line after that heading.

|                 |       |
|-----------------|-------|
| <h1> Hello</h1> | Hello |
| <h2> Hello</h2> | Hello |
| <h3> Hello</h3> | Hello |
| <h4> Hello</h4> | Hello |
| <h5> Hello</h5> | Hello |
| <h6> Hello</h6> | Hello |

**Paragraph Tag <p>:** The <p> tag offers a way to structure your text into different paragraphs. Each paragraph of text should go in between an opening <p> and a closing </p> tag.

<p>This is a paragraph.</p> This is a paragraph.

## HTML Grouping Using <div> and <span>

- **<div> (Division Tag):** A block-level element used to group multiple elements together for styling or layout purposes.
- **<span> (Span Tag):** An inline element used to apply styles or scripts to a specific portion of text within a block.

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## HTML Lists

Lists help organize content in a structured manner. They enhance readability and improve the presentation of information. Lists are widely used in navigation menus, feature lists, instructions, and more.

### Types of HTML Lists

- **Ordered Lists (<ol>):** Used when the order of items is important.
- **Unordered Lists (<ul>):** Used when order does not matter.
- **Definition Lists (<dl>):** Used for presenting terms and their descriptions.

### Ordered Lists (<ol>)

## HTML Tables

HTML tables are used to display **structured data** in rows and columns. They are commonly used for presenting **tabular data**, such as schedules, price lists, and reports.

A table in HTML is created using the <table> element. Inside a table, we use: • <tr> for **table rows**

- <td> for **table data (cells)**
- <th> for **table headers** (bold & centered by default)

## Merging Cells (Colspan & Rowspan)

You can **merge cells** using:

- colspan (merges columns)
- rowspan (merges rows)

## Example: Using colspan

| Personal Details |          |
|------------------|----------|
| Name             | Mohammed |

```
<table border="1">
  <tr>
    <th colspan="2">Personal Details</th>
  </tr>
  <tr>
    <td>Name</td>
    <td>Mohammed</td>
  </tr>
</table>
```

## Example: Using rowspan

```
<table border="1">
  <tr>
    <th rowspan="2">Day</th>
    <th>Morning</th>
    <th>Evening</th>
  </tr>
  <tr>
    <td>Exercise</td>
    <td>Study</td>
  </tr>
</table>
```

| Day | Morning  | Evening |
|-----|----------|---------|
|     | Exercise | Study   |

## Nested Tables

A table inside another table is called a **nested table**.

```
<table border="1">
  <tr>
    <td>Main Table, Row 1</td>
    <td>
      <table border="1">
        <tr>
          <td>Nested Table, Row 1</td>
        </tr>
      </table>
    </td>
  </tr>
</table>
```